

waste land policy 2019. AGEL's proven project execution capabilities, technological prowess, operational excellence and experience of building and operating ultra largescale renewable energy projects, such as India's first and the world's largest wind-solar hybrid cluster and the largest single-location solar park in Jaisalmer, gives it the competitive advantage to build yet another engineering marvel. Through this, the company will play a key role in accelerating India's clean energy transition.

AGEL is developing several innovative solutions that are being piloted at Khavda, providing a scalable blueprint for largescale RE parks across the globe. To enable maximum power output and to reduce the Levelized Cost of Energy (LCOE), it is also deploying advanced renewable technologies and digitisation at the world's largest RE park. A few of the technological innovations are listed below:

- The wind turbine generators (WTGs) with 5.2 MW capacity that are to be deployed here are India's largest and one of the world's most powerful onshore turbines designed specially to suit the wind regime at the location to harness maximum wind for higher energy yield.
- Bifacial solar photo voltaic (PV) modules and horizontal single-axis tracker (HSAT) systems are being used to enable maximum electricity generation from solar energy. Bi-facial modules enable energy generation from both sides of the module

since solar cells are present on both sides.

- TopCon (Tunnel Oxide Passivated Contact) N-type solar cells are being used to convert more sunlight, resulting in a higher cell and module efficiency.
- The Adani Group's Energy Network Operation Centre (ENOC) platform's anchored-in cloud technologies will aid in achieving superior and consistent operational performance of the entire site.
- Waterless cleaning robots will be deployed to counter dust accumulation on solar panels and addresses concerns about water scarcity in the arid Kutch region. This innovation aligns with the UN Sustainable Development Goal 6 (SDG 6).
- A safety kiosk has been deployed at the site to provide the employees with essential self-learning and immersive safety training through the use of virtual reality.
- Three civil labs have been established to carry out field testing for all types of material, concrete cube samples and design mixes to save time with less manual intervention for better quality control.
- To optimise land use, the park is designed in a way wherein each block of 500 MW has both solar panels and WTG co-located.
- A web-based Integrated Project Management System (IPMS) has been implemented to optimise utilisation of resources, planning, executing, monitoring, controlling, real-time communication and systematic data management of the overall project activities.
- Aerial imagery analytics are being used

to monitor the project's progress remotely. This improves project planning and engineering design through topographical and hydrology surveys, and helps in identifying project risks and taking mitigation action through construction monitoring.

- Corrosion resistant coatings and suitable material for solar module mounting structures and switchyard equipment are being used for long-term protection of the plant and equipment from the highly corrosive environment at Khavda.
- Customized geo-tagging solutions are being employed to enable precise location tracking for stores and delivery points within the site to minimise material handling complexities.
- Customised re-usable templates are being utilised for stub foundation to increase the execution speed.
- The project is expected to create over 15,200 green jobs.

Work at Khavda continues at a fast pace, with AGEL leveraging the project execution capabilities of Adani Infra, the manufacturing expertise of Adani New Industries Limited, the operational excellence of Adani Infrastructure Management Services Ltd. and the robust supply chain of its strategic vendors.

The world's biggest RE plant is a testament to AGEL's sustainable progress and unwavering resolve to accelerate India's clean energy transition and enhance social and natural capital in the region.

